Circuit Lab, Division B

Practice Test 2 (State level) · written test · 50 minutes · 50 questions

Team name: _____ Team number: _____

Names: _____

Instructions

- Write all answers on the answer sheet at the back. Only the answer sheet is graded.
- Give numeric answers in metric (SI) units with the correct number of significant figures.
- You may use two calculators and your own paper notes.
- Use $k = 8.99 \times 10^{9}\ \text{N·m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19}\ \text{C}$. Wires and meters are ideal unless the question says otherwise.
- In logic questions, A′ means NOT A, AB means A AND B, and A + B means A OR B.

Section	Questions
Section I: Multiple choice	1–16
Section II: Short answer and calculations	17–36
Section III: Circuit problems	37–50

Section I: Multiple choice

Circle the one best answer. 1 point each.

1. Gustav Kirchhoff first stated his circuit laws in 1845. What was he at that time?
   A. A retired professor
   B. A telegraph engineer
   C. An instrument maker
   D. A university student

2. Which instrument did Coulomb use to measure the force between two charged spheres?
   A. A gold-leaf electroscope
   B. A torsion balance
   C. A galvanometer
   D. A spark gap

3. Galvani said a frog's leg twitched because of "animal electricity". What did Volta argue?
   A. The electricity came from the frog's brain
   B. The electricity came from two different metals in contact
   C. The twitch was caused by heat
   D. Electricity cannot move through living tissue

4. Heinrich Hertz is best known for which discovery (1886-1888)?
   A. The law of electric force
   B. Making and detecting electromagnetic (radio) waves
   C. The first electric battery
   D. The algebra of logic

5. A negative rod is brought NEAR (not touching) the knob of a neutral electroscope. The leaves:
   A. Spread apart, because they become positive
   B. Stay closed, because the electroscope is neutral
   C. Spread apart, because they become negative
   D. Close, because the knob becomes negative

6. Extra charge is placed on a solid metal sphere. When the charges stop moving, where is the extra charge?
   A. Spread evenly through the volume
   B. At the center
   C. It leaks away at once
   D. On the outer surface

7. Why is 60 Hz AC often more dangerous to touch than DC of the same voltage?
   A. AC has a higher resistance
   B. AC makes muscles clamp, so the person cannot let go
   C. AC always carries more charge
   D. DC cannot flow through the body

8. Two identical 1.5 V, 2000 mAh cells are connected in PARALLEL. The pair gives:
   A. 1.5 V and 4000 mAh
   B. 3.0 V and 2000 mAh
   C. 3.0 V and 4000 mAh
   D. 1.5 V and 2000 mAh

9. A voltmeter with an internal resistance of only 10 kΩ is used across a 1 MΩ resistor in a circuit. Compared with the true voltage, the reading is:

A. Much too high
B. Correct
C. Exactly zero
D. Much too low

10. What is one advantage of a circuit breaker over a fuse?

A. It can be reset after it trips
B. It reacts to smaller leakage currents to ground
C. It needs no current rating
D. It changes AC to DC

11. In which case will a GFCI NOT trip?

A. A person standing on wet ground touches the hot wire
B. A person on a rubber mat touches the hot and the neutral wire at the same time
C. A hair dryer falls into a sink and current leaks to the drain pipe
D. The TEST button is pressed

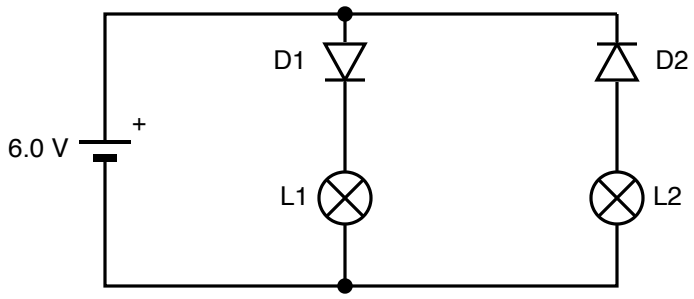
12. A lamp is controlled by two 3-way switches A and B. As a logic function of the switch positions, the lamp is best described as:

A. AND
B. XOR (or XNOR)
C. OR
D. NAND

13. Silicon is doped with phosphorus. What kind of semiconductor results, and what are its majority carriers?

A. n-type; electrons
B. p-type; holes
C. p-type; electrons
D. n-type; holes

14. In the circuit below, which lamp or lamps light? The diodes are silicon. The battery's + side is at the top.



A. L2 only
B. Both L1 and L2
C. Neither lamp
D. L1 only

15. A half adder adds bits A and B. Which gates make its Sum and Carry outputs?

A. Sum = XOR, Carry = AND
B. Sum = OR, Carry = AND
C. Sum = AND, Carry = XOR
D. Sum = XNOR, Carry = NOR

16. What is the smallest number of 2-input NAND gates that can make a 2-input OR gate?

A. 1 **B.** 2

C. 3 **D.** 4

Section II: Short answer and calculations

Write the answer with units and the correct number of significant figures. 2 points each.

17. A +2.5 μC charge and a −4.0 μC charge are 15 cm apart. Find the size of the force between them and say whether it is attractive or repulsive.

18. Three point charges lie on the x-axis: +3.0 μC at x = 0, −2.0 μC at x = 0.30 m, and +1.0 μC at x = 0.60 m. Find the size and direction of the net force on the −2.0 μC charge.

19. Find the electric field and the electric potential 0.25 m from a +6.0 nC point charge.

20. A battery has an EMF of 9.0 V. When it supplies 0.30 A, its terminal voltage is 8.1 V. Find its internal resistance.

21. A European outlet has a peak voltage of 325 V. What is its rms voltage?

22. Nichrome has $\rho = 1.10 \times 10^{-6}$ Ω·m. What length of 0.50 mm diameter nichrome wire has a resistance of 12.0 Ω?

23. A 2.0 kW heater runs for 45 minutes. How much energy does it use, in joules and in kWh?

24. A 40 W bulb and a 100 W bulb (both rated for 120 V) are connected in SERIES across 120 V. Find the power used by the 40 W bulb. Which bulb is brighter?

25. A 5-band resistor reads orange, white, black, brown, brown. Give its value and tolerance.

26. A Wheatstone bridge has R1 = 120 Ω (top left), R2 = 360 Ω (bottom left), R3 = 150 Ω (top right) and the unknown R_x (bottom right). The meter across the middle reads zero. Find R_x.

27. Two 100 kΩ resistors are in series across 10.0 V. A voltmeter with 1.0 MΩ internal resistance is placed across the bottom resistor. What does it read? (Without the meter the voltage is 5.00 V.)

28. A 5.0 V battery, a forward-biased silicon diode (0.7 V) and a 220 Ω resistor are in series. Find the current.

29. Two red LEDs (2.0 V each) are in series with one resistor on a 9.0 V battery. What resistor gives 20 mA?

30. Simplify F = A′B′C + A′BC + AB′C + ABC′ + ABC.

31. For inputs ABC = 000, 001, ..., 111 (in order), the output column is 0, 1, 1, 0, 1, 0, 0, 1. Write F as a sum of products. What single gate does this?

32. Add the binary numbers 101101 + 11011. Give the sum in binary and in decimal.

33. A 2-to-1 multiplexer has data inputs I_0 and I_1 and a select input S. Output Y = I_0 when S = 0 and Y = I_1 when S = 1. Write Y as a Boolean expression.

34. In a hands-on task, a 9.0 V supply drives a resistor network. The ammeter in series with the supply reads 3.6 mA. What power does the supply deliver?

35. You have one each of 220 Ω, 330 Ω, 470 Ω and 1.0 kΩ. Using all four, make a resistance as close to 400 Ω as you can. Give the arrangement and its resistance.

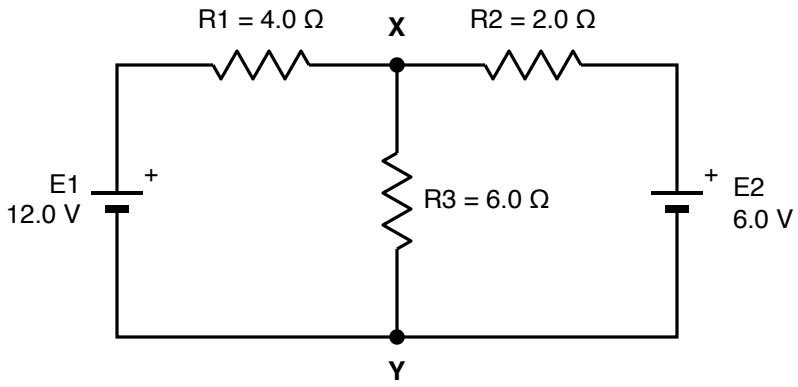
36. A red LED (2.0 V) and a green LED (3.0 V) are in separate parallel branches on a 9.0 V battery, each with its own resistor. Both LEDs must carry 15 mA so they look equally bright. Find the resistor for the GREEN LED.

Section III: Circuit problems

Show your work. 2 points per part.

Problem A: two batteries (Kirchhoff's laws) (Questions 37–42)

E1 = 12.0 V, E2 = 6.0 V, R1 = 4.0 Ω, R2 = 2.0 Ω, R3 = 6.0 Ω. Both batteries have their + side at the top. Junction X is at the top of R3 and junction Y at the bottom. Ideal batteries and wires. Let I_a be the clockwise current in the left loop and I_b the clockwise current in the right loop.



37. Write the two loop (KVL) equations in I_a and I_b.

38. Find the current through R1.

39. Find the current through E2. Is E2 being charged or discharged?

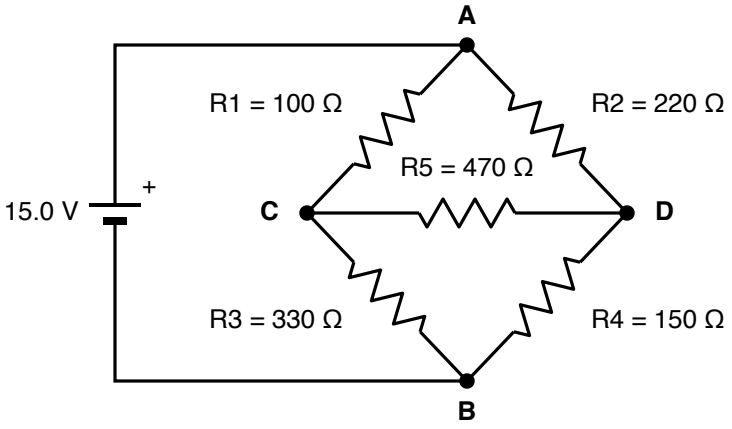
40. Find the current through R3 and its direction.

41. Find the voltage of junction X relative to junction Y.

42. Find the power delivered by E1.

Problem B: unbalanced bridge (Questions 43–45)

A 15.0 V battery is connected between A (+) and B (–). R1 = 100 Ω (A-C), R2 = 220 Ω (A-D), R3 = 330 Ω (C-B), R4 = 150 Ω (D-B), R5 = 470 Ω (C-D). The bridge is not balanced (R1/R3 is not equal to R2/R4).



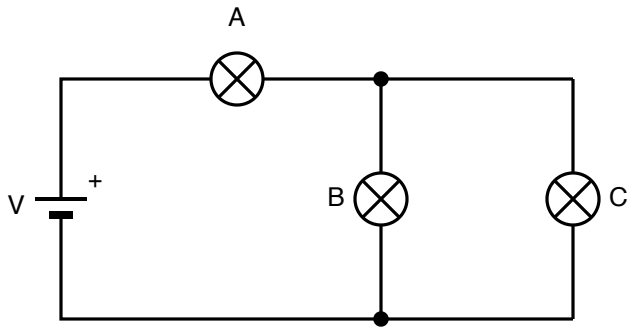
43. Find the voltage across R5 ($V_C - V_D$). Which end is higher?

44. Find the total current from the battery.

45. Find the equivalent resistance between A and B.

Problem C: three identical bulbs (Questions 46–48)

Three identical bulbs A, B and C are connected to an ideal battery as shown. Treat each bulb as a fixed resistor R. Brightness follows power.



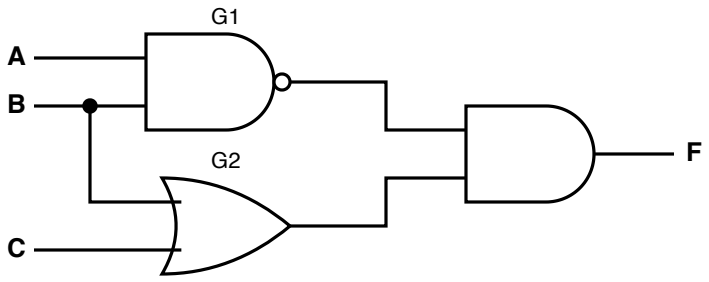
46. Bulb C is unscrewed (removed). By what factor does the power of bulb A change? Does A get brighter or dimmer?

47. With C still removed, by what factor does the power of bulb B change compared with the start?

48. Instead, C is put back and a wire is connected across bulb A (A is shorted). What happens to each bulb?

Problem D: logic circuit (Questions 49–50)

Gate G1 is a NAND gate with inputs A and B. Gate G2 is an OR gate with inputs B and C. An AND gate combines G1 and G2 to make F.



49. Write F and simplify it to a sum of two products.

50. List every input row ABC that makes F = 1.

End of the test. Check your answer sheet.

Answer sheet · Practice Test 2 (State level)

Team name: _____ Team number: _____

Multiple choice: circle one letter. Other questions: write the answer with units.

1. A B C D
2. A B C D
3. A B C D
4. A B C D
5. A B C D
6. A B C D
7. A B C D
8. A B C D
9. A B C D
10. A B C D
11. A B C D
12. A B C D
13. A B C D
14. A B C D
15. A B C D
16. A B C D
17. _____
18. _____
19. _____
20. _____
21. _____
22. _____
23. _____
24. _____
25. _____
26. _____
27. _____
28. _____
29. _____
30. _____
31. _____
32. _____
33. _____
34. _____
35. _____
36. _____
37. _____
38. _____
39. _____
40. _____
41. _____
42. _____
43. _____
44. _____
45. _____
46. _____
47. _____
48. _____
49. _____
50. _____